DATABASE MANAGEMENT SYSTEMS LABORATORY
Course Code: 06123228
Lab Test: 02
Full Marks: 20
Time: 60 Minutes

Objective
Practice the use of different SQL joins:

- INNER JOIN
- LEFT JOIN
- RIGHT JOIN
- FULL JOIN
- SELF JOIN

Note: MySQL/MariaDB used with XAMPP does not directly support FULL OUTER JOIN. Students can demonstrate the equivalent result using LEFT JOIN + RIGHT JOIN with UNION.

Problem Statement
A university maintains information about **students, departments, and instructors** using the following tables.

Table 1: Department

DepartmentID	DepartmentName
1	CSE
2	EEE
3	BBA
4	Civil

Table 2: Student

StudentID	StudentName	DepartmentID
101	Rahim	1
102	Karim	2
103	Nusrat	1
104	Rafi	3
105	Sadia	1

Table 3: Instructor

InstructorID	InstructorName	DepartmentID	SupervisorID
201	Dr. Ahmed	1	NULL
202	Dr. Karim	1	201
203	Dr. Hasan	2	201
204	Dr. Rahman	3	201
205	Dr. Nayeem	4	NULL

Tasks

Task 1 — INNER JOIN
Display the **student name and department name** for every student who belongs to a department.
Expected columns:
StudentName | DepartmentName

Task 2 — LEFT JOIN
Display **all departments and their students**, including departments that currently have **no students**.
Expected columns:
DepartmentName | StudentName

Question: Which department appears without a student?

Task 3 — RIGHT JOIN
Using a RIGHT JOIN, display **all departments and their students**, including departments that have no students.
DepartmentName | StudentName
****Compare the result with Task 2.**

Task 4 — FULL JOIN
Display **all students and all departments**, including unmatched records.
Since XAMPP uses MySQL/MariaDB, implement the FULL OUTER JOIN result using:
LEFT JOIN
UNION
RIGHT JOIN

Task 5 — SELF JOIN
The Instructor table contains a SupervisorID that refers to another instructor.
Display:
InstructorName | SupervisorName

For example:
Dr. Karim | Dr. Ahmed
Dr. Hasan | Dr. Ahmed

Hint: Join the Instructor table with itself.

Task 6 — Three-Table JOIN
Display each student's:
StudentName
DepartmentName

using the Student and Department tables.
Then extend the query to include instructor information:
StudentName | DepartmentName | InstructorName

Task 7 — Challenge
Find the departments that **do not have any students**.
Expected Output:
Civil

Submission Requirements
Students should submit:

1. SQL statements for Tasks 1–7.
2. Screenshots of each query and its output.
3. A short explanation of the difference between:
    - INNER JOIN
    - LEFT JOIN
    - RIGHT JOIN
    - FULL JOIN
    - SELF JOIN

Suggested Marks: 20

Task	Marks
INNER JOIN	3
LEFT JOIN	3
RIGHT JOIN	3

FULL JOIN	4
SELF JOIN	3
Three-table JOIN	2
Challenge	2
Total	**20**